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Submitter Information

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General Comment

Suggestions for AI security research (from a professional security researcher, wannabe mathematician and ai security reseracher)

I love that you're doing this.
And it would be amazing if us firms weren't so hamstrung by copyright law. Like why is some legacy magazine able to sue a leading industry over accidentally consuming their news articles or not paywalling their news articles effectively and losing 'valuable content' which is worth nothing.

Although civil law has long been a regulating factor it has also notably been weaponized by banks and other industries to stymie innovation, oppress and censor individuals and prevent non-competitive research and ideas.

However that's not my specific domain and not what I'd like to discuss.

Security research since I got into it has been the most important field. In 2006 when I was young I knew it was important because all the money in the world was regulated by the SWIFT system which was protected by cryptographic protocols. I knew this because I was fortunate, for I had family who worked in finance.

I thought, what could be a more competitive and interesting industry than that which managed all the value in the world.

I was not wrong... and the industry has grown and it has grown. With it there has been inovation and now with XBOW and other companies doing wonderful things juxtaposed with AI the darpa grand cyber challenge being solved, in an innovativeless way. It is clear that we are entering the era of ICE not softice the kernel debugger but BLACK ICE like from sprawl series.

We are looking at fully automated attack (and defense) pipelines.

We as the USA should be on the forefront of this. As a patriot and an american I know that it can be hard to push for the right thing in politics but I also know that this is essential to maintaining information and technological advantages.

Given all this I suggest the following two key points:
1) Increase penalties for selling weaponized ai security technology to foreign assets (I know this is a strange suggestion from a hacker from the bay area but even when I was very young.. I knew we had crypto embargos and for good reason. It is sane to realize that this has the potential to be more valuable in some ways to military communication security than many cryptographic advances and it needs to be protected very closely to if not at the same tier.

2) Remove penalties and risks for US security researchers
If I want to say learn more about AI security I currently need to use platforms like crucible. But really I should be able to try and extract models from our major commerical providers. This is because if they cannot secure their model (i.e. deepseek) this creates a security risk for the USA. We should encourage innovation in this manner if Y compkany can't do it someone else can.... SMALL CAVEAT: I do not think researchers should be able to profit from this. Us few intellectuals who do this smashing for fun and not profit should not be hounded civilally, criminally or nationally. We should be encouraged to help move things forward. We don't want another deepseek fiasco do we?

I will elaborate further in addition to extracting models from blackboxes we should permit: doing CoT analysis, masking and manipulation of models(obviously jailbreaking), developing novel offensive cyber weapons using ML (god I don't like the term cyber and never will) and anything across the rainbow unicorn sky of AI weaponry. My list is replete and I know it and you should too, carve out buffers for the future encouragement of us pushing the envelope.

I would suggest the following radical mantra:
SECURITY FIRST, ECONOMIC DOLLARS second. Or do I need to reiterate how SWIFT works?

We should allow security researchers and private entities to hack and hack and hack an AI security project they want that doesn't cause loss of life or ecnomic resources intentionally even though mistakes will happen

We should to the maximum ability of the US government encourage researchers to do thusly, protect them from civil and criminal persecution and provide resources and education to indivduals, programs sponsoship whatever it takes to get us ahead of the curve

As a student of the world having living in china for a while when I was young I know that china has been heavily focused for the past couple decades on AI and math training and I know that from a human resource perspective we are behind(anyone else read arxiv?). When I was younger in 2012 my ML book for an actual course was the mandarin international edition, which was very cheap and printed many times.

We need to get ahead, we need to innovate, if we choose to stymie research for ephermeral momentary corporate dollars we will have deepseek after deepseek after deepseek until we are on the other end. Please do not go the route of DMCA with DRM researchers for AI security because then I will have to move and the USA will be absolutely ruined.